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Stronger renewal in human fear conditioning when tested with an acquisition retrieval cue than with an extinction retrieval cue

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Abstract

It was previously demonstrated in our laboratory that conditioned fear in humans can renew after extinction, when this procedure took place in a different context (ABA-renewal [Vansteenwegen et al. (2005). Return of fear in a human differential conditioning paradigm caused by a return to the original acquisition context. *Behaviour Research and Therapy, 43*(3), 323–336]). Using the same experimental design, we now tested the power of retrieval cues to interact with this contextual renewal phenomenon. Two groups went through acquisition and extinction (in a different context). They were then tested in the original acquisition context and in the presence of a retrieval cue. In the acquisition-cue group, this cue previously featured during the acquisition procedure; in the extinction-cue group, the cue previously featured during the extinction procedure. As expected, renewal of conditioned electro-dermal responding and retrospective expectancy ratings was strongest in the acquisition-cue group. Theoretical and clinical implications of this finding are discussed.
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Introduction

Background contexts play an important role in the behavioural expression of acquired knowledge. For instance, human memory research has demonstrated that information is better retrieved when the recall test takes place in the learning context (e.g. Smith & Vela, 2001). This observation is in accordance with some animal conditioning studies, where the conditional response (CR)—resulting from paired presentations of a conditional stimulus (CS) with an unconditional stimulus (US)—is smaller when the CS is presented in a different context (Bonardi, Honey, & Hall, 1990; Hall & Honey, 1990; Lovibond, Preston, & Mackintosh, 1984). Furthermore, the impact of background contexts in animal conditioning preparations appears to be even larger in extinction performance than in acquisition performance (Bouton, 1993, 1994; Bouton & Moody, 2004). In extinction, an animal is exposed to CS-only presentations. Typically, the previously acquired CR

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decreases gradually. Numerous studies in the animal conditioning literature have demonstrated that a subsequent change of context largely disrupts extinction performance. The result is renewed responding to the CS (Bouton & Bolles, 1979; Bouton & King, 1983; Bouton & Ricker, 1994). Interestingly, an analogous effect is demonstrated in human causal learning (which is on the verge between animal conditioning and human memory experiments), where second-learned information to the same cue is found to be especially context-dependent (Garcia-Gutierrez & Rosas, 2003).

Vansteenwegen et al. (2005) demonstrated the context-dependency effect of extinction in a human autonomic conditioning preparation. Here, the picture of a neutral face serves as CS and is repeatedly followed by the administration of an aversively loud noise (i.e., the US). The dependent variable (CR) is the on-line measurement of the skin conductance response (SCR), which is related to expectancy (of the aversive event) and action tendency. As such, it is often taken as an index of fear. Because the SCR can also reflect orienting and habituation effects, a control stimulus is normally added to the conditioning procedure. This stimulus (CS^-) goes through the same experimental procedure as the experimental stimulus (CS^+) but is never paired with the US. The CR is then defined as the differential response to the CS^+ and the CS^-. During extinction, both stimuli are repeatedly presented without the US. Crucially, extinction is conducted in a different context (provided by switching out/on the houselight). Vansteenwegen et al. (2005) observed that a return to the acquisition context, after complete extinction of the CR, is sufficient to renew this CR to the CS^+. In addition, retrospective expectancies of the shock rated upon completion of the experiment also demonstrated the typical renewal effect.

Renewal demonstrations of extinguished fear responding have potential relevance for the treatment of anxiety disorders (see Bouton, 2002), as they point to context-dependency of fear extinction as a possible determinant for the sometimes reported return of fear after successful therapy completion (see Rachman, 1989). In exposure therapy, the context may be provided by the physical context in which exposure is performed or the presence of the therapist (Mineka, Mystkowski, Hladek, & Rodriguez, 1999; Rodriguez, Craske, Mineka, & Hladek, 1999), perceptual features of the exposure stimuli (Rowe & Craske, 1998; Vervliet, Vansteenwegen, Baeyens, Hermans, & Eelen, 2005; Vervliet, Vansteenwegen, & Eelen, 2004) or the presence of concurrent stimuli (Lovibond, Davis, & O'Flaherty, 2000).

Different strategies to reduce the renewal effect have been suggested and approved in the animal conditioning literature as well as in clinical populations (e.g. Brooks & Bouton, 1994; Chelonis, Calton, Hart, & Schachtman, 1999; Gunther, Denniston, & Miller, 1998; Rowe & Craske, 1998). Collins and Brandon (2002) demonstrated less renewal of craving for alcohol cues when during test cues were presented that were also present during extinction. Similarily, Mystkowski, Echiverri, Labus, and Craske (in press) showed less effects of context-change after exposure when spider fearful participants were asked to mentally reinstate the treatment context. In an elegant animal conditioning preparation, Brooks and Bouton (1994) also demonstrated the attenuation of renewal by the use of retrieval cues. They introduced discrete cues during the extinction phase of a standard ABA-renewal procedure. The cues repeatedly preceded the non-reinforced CS trials during this phase. Interestingly, in the test phase in the original acquisition context the CS caused *less* renewal when it was preceded by presentations of the extra cue then when it was tested alone.

Inspired by this last study, we tested whether the introduction of an extra discrete cue can interact with the ABA-renewal effect in human fear conditioning. The basic ABA-renewal procedure was identical to Vansteenwegen et al. (2005). Two groups were compared. The extinction-cue group (EC) received presentations of the cue during extinction and test. The acquisition-cue group (AC) received presentations of the cue during acquisition and test. The predictions were straightforward: If the extra cues have the power to interact with the renewal effect, the test response should be stronger in AC as compared to EC.

Method

Participants

Thirty-two first-year psychology students participated in order to fulfill course requirements. Sixteen participants were assigned to the AC and 16 to the EC. They all gave informed consent and were informed that they could decline to participate at any time.

Apparatus

Two clearly distinct line drawings of pictorial faces served as conditioned stimuli. For half of the participants, one picture was followed by the US (CS^+) and the other stimulus was not (CS^-). For the other half of the participants this assignment was reversed. A slide depicting a small black cross in the middle served as cue-stimulus. The slides were projected with a Kodak Carousel slide projector and shutter, controlled by Labtech Notebook programme and an IBM computer, which also controlled the stimulus sequence, the presentation duration and the intertrial intervals. Slides were projected at eye level at 40 cm by 60 cm. A 95 dB (A) (Bruël and Kjaer, type 2107) burst of white noise with instantaneous rise time presented binaurally for 500 ms through headphones (Sony dynamic stereo, MDR-CD270), served as US Electro-dermal activity was recorded with Fukuda standard Ag/AgCl electrodes (1 cm diameter) filled with a Unibase electrolyte and attached to the hypothenar palm of the left hand which was cleaned with tap water. The inter-electrode distance was 2.5 cm. The Coulbourn skin conductance coupler (S71-22) provided a constant 0.5 V across electrodes. The analog signal was passed through a 12-bit AD-converter and digitized at 10 Hz from 7 s prior to conditional stimulus onset until 4 s after conditional stimulus offset.

Participants were seated in an armchair (screen distance 1.5 m) in a sound attenuated experimental room, adjacent to the experimenter's room. Verbal communication was possible through an intercom system. The experimenter could manipulate the central lighting in the experimental room from the outside.

Procedure

At the beginning of the experiment the participants were instructed that from time to time a loud aversive noise would be presented and that they had to watch the slides attentively. They were instructed to focus on the moments when a loud aversive noise occurred. No further instructions were given with regard to the contingencies between CSs and the US. During acquisition, extinction and test, slides were presented for 8 s. Intertrial intervals varied between 16 and 24 s (average 20 s) and electro-dermal activity was recorded from 7 s before CS onset to 4 s after CS offset. During acquisition, CS^+ and CS^- were presented 10 times in a semi-randomized order. During this phase the CS^+ was always immediately followed by the US. The CS^- was never followed by the US. For the participants from the AC-group 8 out of 10 presentations of the CS^+ and 8 out of 10 presentations of the CS^- were preceded by the cue-stimulus. The cue was presented for 4 s and the interval between cue-offset and CS-onset was 1 s. During extinction, CS^+ and CS^- were again presented 10 times each, but now both without the US, in a semi-randomized order. For the participants from the EC-group 8 out of 10 presentations of the CS^+ and 8 out of 10 presentations of the CS^- were preceded by the cue-stimulus during extinction. The cue was presented for 4 s and the interval between cue-offset and CS-onset was 1 s. Finally during test, CS^+ and CS^- were presented three times each, in a semi-randomized order. All CSs in both experimental groups were preceded by the cue-stimulus during the test phase. For all participants, extinction was conducted in a different context than acquisition and test. For half of the participants, acquisition and test occurred with the central lighting in the experimental room switched off, whereas during extinction the lighting was switched on. For the other half of the participants, acquisition and test took place in a lightened room whereas for extinction the central lighting was switched off.

After the test phase, the recording electrodes were removed and the participants were taken to the adjacent experimenter's room. The participants were asked to complete a graph representing the evolution of their US-expectancies during the just finished experiment. The experimenter explained the meaning of the two axes. The *Y*-axis of the graph depicted an 11-point US-expectancy scale starting at 0 (never) and ending at 10 (always). On the *X*-axis, three phases were indicated. The first phase was described as before it became dark (/light), the second as when it was dark (/light) and the third as when it became light (/dark) again. For each phase the participants were asked to indicate twice as to what extent they expected the US: once at the beginning of that phase and once at the end of that phase. After the instructions, the participants of both groups saw a picture of the CS^+ and were asked to give the six ratings for the CS^+. Then they were asked to do the same for the CS^-. Afterwards participants had to rate the pleasantness of the US on an 11-point graphic scale (anchored: -10, *unpleasant* and $+10$, *pleasant*).

Results

US-ratings

The mean pleasantness rating for the US at the end of the experiment was -5.87 (SD $= 2.77$) in the EC-group and -4.62 (SD $= 3.32$) in the AC-group. There was no significant difference between the two groups, $F(1,30) = 1.33$, MSE $= 9.38$, $p = 0.26$.

Retrospective US-expectancy ratings

In Fig. 1 mean US-expectancy ratings for CS^+ and CS^- for the six retrospective test moments are presented, for the AC-group and the EC-group separately. The graph demonstrates clear acquisition and extinction in both groups, as well as renewed differential expectancy to the CS^+ and CS^- at test. Crucially, the renewal effect is more pronounced in the AC-group. Statistical tests confirm these conclusions.

US-expectancy for CS^+ and CS^- for the six moments and for each group were analysed in a Group (AC, EC) $\times$ Moment (acq1, acq2, ext1, ext2, test1, test2) $\times$ CS-type (CS^+, CS^-) ANOVA with repeated measurements on the two last variables.

Acquisition and extinction: One-degree-of-freedom contrasts confirm that both groups show acquisition. There was no differentiation between CS^+ and CS^- at the beginning of the first phase, AC: $F(1,30) < 1$; EC: $F(1,30) = 2.35$, $p = 0.14$, MSE $= 2.99$, but a clear differentiation at the end in both groups, AC: $F(1,30) = 240$, $p < 0.01$, MSE $= 2.77$; EC: $F(1,30) = 196$, $p < 0.01$, MSE $= 2.77$. The differentiation at the end of acquisition between the two groups was not significantly different, $F(1,30) = 1.10$, $p = 0.30$, MSE $= 2.78$. During the extinction phase the differentiation between CS^+ and CS^- declined. For the AC-group, the differentiation at the beginning of extinction was significant, $F(1,30) = 0.81$, $p < 0.01$,

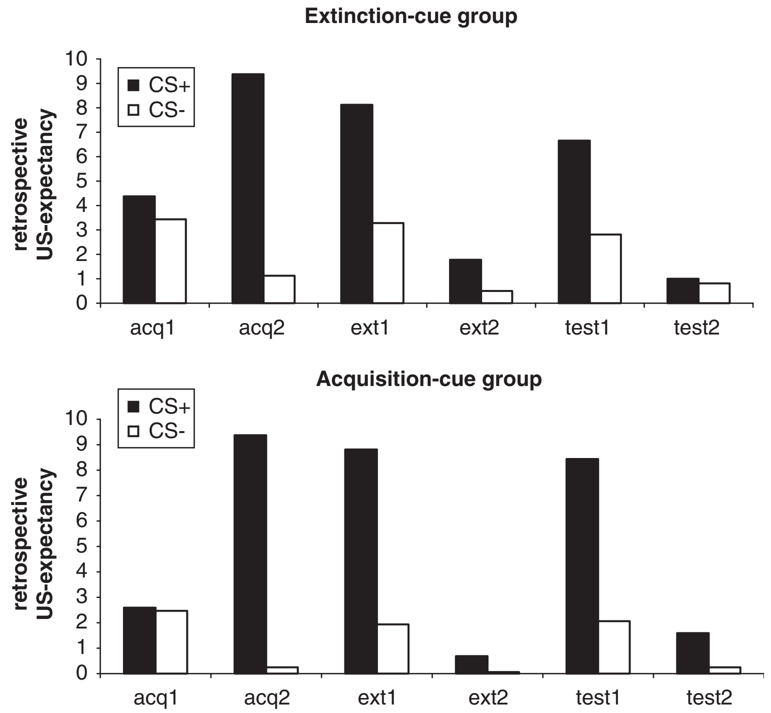

Fig. 1. Mean retrospective US-expectancy ratings are presented for CS^+ and CS^- during six moments separately for the acquisition-cue group ($n = 16$) and the extinction-cue group ($n = 16$) at six moments: beginning of acquisition (acq1), end of acquisition (acq2), beginning of extinction (ext1), end of extinction (ext2, beginning of test (test1) and end of test (test2).

$\mathrm{MSE} = 4.66$but it was not at the end of extinction, $F(1, 30) = 1.76$, $p = 0.21$, $\mathrm{MSE} = 1.86$. For the EC-group a differentiation was obtained at the beginning of extinction, $F(1, 30) = 0.40$, $p < 0.01$, $\mathrm{MSE} = 4.66$ as well as at the end $F(1, 30) = 7.05$, $p = 0.01$, $\mathrm{MSE} = 1.86$. This indicates that in the EC-group the differentiation was not completely lost throughout extinction. However, the Group X CS-type interaction at the end of extinction was not significant, $F(1, 30) < 1$. For both groups, the decline from the beginning to the end of extinction was also significant, AC: $F(1, 30) = 50$, $p < 0.01$, $\mathrm{MSE} = 3.12$; EC: $F(1, 30) = 16$, $p < 0.001$, $\mathrm{MSE} = 3.12$.

Renewal of conditioned responding: For the first rating of the test phase, the AC-group showed a significant differentiation between CS^+ and CS^-, $F(1, 30) = 54$, $p < 0.001$, $\mathrm{MSE} = 6.04$. Ratings for the CS^+ were larger for the first test rating than for the last extinction rating, $F(1, 30) = 127.43$, $p < 0.01$, $\mathrm{MSE} = 3.77$, and the CS^+/CS^- differentiation at the first test rating was significantly larger than the differentiation at the last extinction rating, $F(1, 30) = 42.27$, $p < 0.01$, $\mathrm{MSE} = 3.13$. All these data point to a renewal-effect in this group.

In the EC-group a very similar data pattern was obtained. For the first rating of the test phase, the EC-group showed a significant differentiation between CS^+ and CS^-, $F(1, 30) = 19.5$, $p < 0.01$, $\mathrm{MSE} = 6.04$. Ratings for the CS^+ were larger for the first test rating than for the last extinction rating, $F(1, 30) = 50.42$, $p < 0.01$, $\mathrm{MSE} = 3.77$ and the CS^+/CS^- differentiation at the first test rating was significantly larger than the differentiation at the last extinction rating, $F(1, 30) = 8.39$, $p = 0.01$, $\mathrm{MSE} = 3.13$. Hence also in this group renewal was observed.

When we compare the two groups, data seem to corroborate our predictions. The renewal effect was more pronounced in the AC-group than in the EC-group. The differentiation during the first test rating was larger in the AC-group than in the EC-group, $F(1, 30) = 4.22$, $p = 0.048$, $\mathrm{MSE} = 6.04$ as well as the increase in differentiation from the last extinction rating towards the first test rating, $F(1, 30) = 6.49$, $p = 0.02$, $\mathrm{MSE} = 3.13$.

Skin conductance

Skin conductance responses were visually inspected and corrected for artefacts before they were analysed statistically. Skin conductance response amplitudes were defined as the maximal increase starting within 1–4 s after conditional stimulus onset. Trials with negative changes were scored as zero and included in all analyses. Amplitudes were range corrected using the largest (unconditioned) response (peak between 9 and 13 s during the acquisition phase) elicited by the loud aversive noise (Lykken & Venables, 1971) as the maximum range for each individual. The corrected response magnitudes were subjected to a square root transformation in order to normalize the distribution prior to statistical analysis.

For each participant, mean amplitudes for CS^+ and CS^- were calculated for five blocks of three trials: the first three (acq1) and the last three acquisition trials (acq2), the first three (ext1) and the last three extinction trials (ext2) and the three test trials (test). These means were analysed with a Group (AC, EC) X Moment (acq1, acq2, ext1, ext2, test) X CS-type (CS^+/CS^-) ANOVA with repeated measurements on the last two variables. This analysis contained all relevant comparisons.

Fig. 2 shows the mean amplitudes for CS^+ and CS^- for the 10 acquisition trials, 10 extinction trials and three test trials per group. The graph demonstrates acquisition and extinction in both groups, although the acquisition effect is strongly reduced by the end of the first phase in group AC. At test, only group AC shows a return of differential responding to the CS^+ and the CS^-. Statistical analyses are in line with these conclusions.

Acquisition and extinction: One-degree-of freedom contrasts confirm that both groups show acquisition. In the first block, responses to CS^+ were larger than to CS^- in both groups, AC: $M_{cs+} = 0.37$, $M_{cs-} = 0.23$, $F(1, 30) = 4.44$, $p = 0.044$, $\mathrm{MSE} = 0.03$; EC: $M_{cs+} = 0.56$, $M_{cs-} = 0.38$, $F(1, 30) = 7.45$, $p = 0.011$, $\mathrm{MSE} = 0.03$. The interaction between the two groups for the first acquisition block was not significant, $F(1, 30) < 1$. At the end of acquisition, there was a significant CS^+/CS^- differentiation in the EC-group, $M_{cs+} = 0.54$, $M_{cs-} = 0.32$, $F(1, 30) = 15.33$, $p < 0.001$, $\mathrm{MSE} = 0.03$, but not in the AC-group, $M_{cs+} = 0.26$, $M_{cs-} = 0.21$, $F(1, 30) < 1$. Nevertheless acquisition in the AC-group clearly took place: Responses to the CS^+ were larger than to the CS^- when looking at the mean responses during the fourth, fifth, sixth and seventh trial of acquisition in this group, $M_{cs+} = 0.39$, $M_{cs-} = 0.22$, $t(15) = 3.15$, $p < 0.01$. Habituation to the shock over conditioning trials is often observed in electro-dermal responding (see Vervliet et al., 2005 for a similar

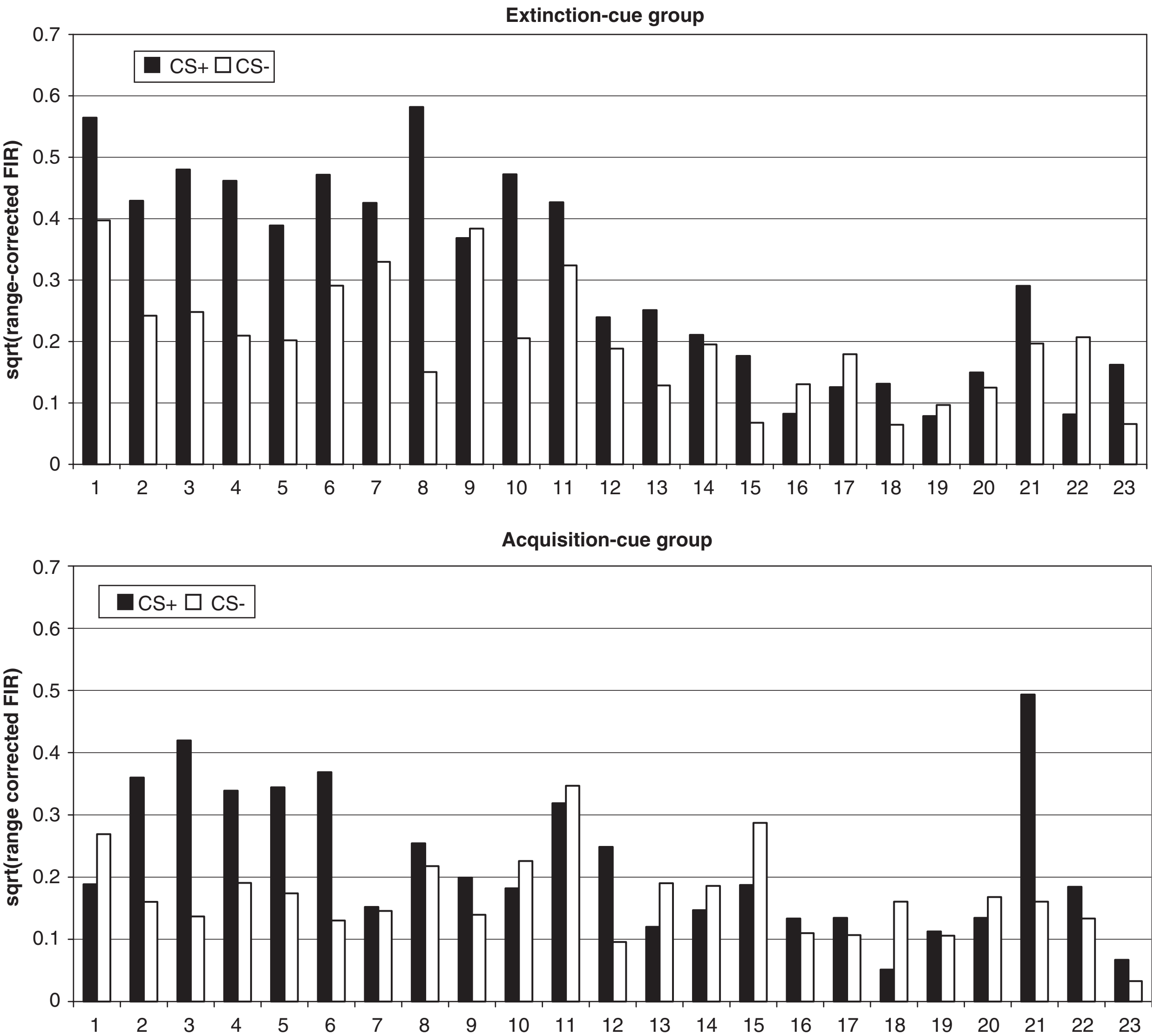

Fig. 2. Mean range-corrected first interval skin conductance response amplitudes during viewing CS^{+} and CS^{-} for 10 acquisition trials (1–10), for 10 extinction trials (11–20) and three test trials (21–23) are presented separately for the acquisition-cue group and the extinction-cue group. A square-root transformation was used in order to obtain normally distributed data.

argumentation). In this case, it is possible that the habituation process was facilitated by the presence of the cues in the AC-group compared to the EC-group.

In both groups, the differentiation between CS^{+} and CS^{-} was already gone in the first extinction block, AC: $M_{cs+} = 0.28$, $M_{cs-} = 0.30$, $F(1,30)<1$; EC: $M_{cs+} = 0.28$, $M_{cs-} = 0.24$, $F(1,30) = 1.49$, $p = 0.23$, MSE $= 0.04$. There was also no significant CS^{+}/CS^{-} differentiation in the last extinction block in both groups, both $F(1,30)<1.01$, AC: $M_{cs+} = 0.17$, $M_{cs-} = 0.13$; EC: $M_{cs+} = 0.13$, $M_{cs-} = 0.18$.

Renewal of conditioned responding: In the AC-group, responses to the CS^{+} were significantly larger during test than during the second extinction block, $F(1,30) = 7.67$, $p = 0.01$, MSE $= 0.04$. Moreover this was not the case for CS^{-}, $F(1,30)<1$. Furthermore, the CS^{+}/CS^{-} differentiation was larger during the test block than during the last extinction block, CS-type X (Ext2, test): $F(1,30) = 4.33$, $p = 0.046$, MSE $= 0.04$. Hence, a return of conditioned responding in the AC-group was obtained.

In the EC-group, the increase for CS^+ from extinction to test and the CS-type X (Ext2, test) interaction were not significant, both $F(1, 30) < 1.72$, so there was no evidence for a return in this group.

When comparing the two between-group effects directly, no significant interactions between the two groups were obtained.

Discussion

The present study demonstrates that the effects of a context change after extinction can be influenced by the presence of a concurrent cue. The experimental history of this extra cue is crucial: We observed more renewal when the cue had previously been presented during acquisition (group AC) as compared to a cue that had previously been presented during extinction (group EC). The retrospective ratings show renewal in both groups, in that the differential shock expectancy increases significantly from extinction to test. This effect of the context change after extinction is in accordance with previous experiments in our lab (Vansteenwegen et al., 2005). The interesting observation is that this renewal effect is significantly stronger in group AC than in group EC. Likewise, the results of the on-line skin conductance measurement show renewal in group AC, but not in group EC. Differential responding to the CS^+ and CS^- increases on the transition from extinction to test in group AC, which is in accordance with earlier demonstrations of renewal in our lab (Vansteenwegen et al., 2005). However in the EC-group, this renewal effect failed to produce an effect.

The US-expectancy-ratings can be criticized because they were taken retrospectively and therefore possibly confounded by the retrieval of the participant's knowledge about the experiment and experimental demands at the moment of test (see Vansteenwegen et al., 2005; Vervliet et al., 2005, for discussions). Some elements speak against these criticisms. First, we obtained recovery-effects despite the fact that the participants in fact experienced no US's in the test phase. This makes clear that the participants were able to interpret the graph and the US-expectancy scale in a proper way. Moreover, it suggests that it is less probable that the ratings exclusively reflected demand-effects. Second, we obtained a different amount of recovery in the two groups that can only be attributed to the action of the extra cue. It is difficult to see how this difference can be explained by demand effects because this would require a rather sophisticated reasoning from the participants. Third and most importantly, the data pattern of the skin conductance data that were taken on-line seem to confirm the obtained effects in the US-expectancy ratings.

However, although a similar data pattern was obtained, the results of the two measures were not *completely* identical. First, group EC demonstrated renewal in the expectancy ratings but not in the skin conductance. Second, the between groups difference concerning the renewal effect was significant in the expectancy ratings but not in the skin conductance. It is a common finding, however, that skin conductance results produce significant higher-order interactions with difficulty, probably because of the high response variability, both inter- and intra-individual (see Öhman, 1983). Hence, as the only differences obtained, are mainly related to statistical power and as the general data pattern for the two measures is highly similar, we can consider these data as converging evidence for our hypothesis. We demonstrated the power of retrieval cues to influence contextual renewal in human fear conditioning.

Note that the experimental design controls for non-associative effects. First, the addition of the control stimulus (CS^-) allows controlling for mere orienting and dishabituation (to which the skin conductance is very sensitive) during the entire course of the experiment. Second, the fact that the extra cue is presented equally frequent in both groups ensures that the obtained results are not due to a difference in familiarity with that cue at test. The only difference is the *moment* in which it was presented during the experiment (acquisition in group AC versus extinction in group EC). Although the larger time gap between the cued phase and the test phase in group AC than in group EC, could have induced larger responses in group AC in general, this difference cannot explain the larger CS^+/CS^- differentiation in this group.

The employed design, however, does not allow identifying the associative mechanism responsible for the obtained results. Although we used a procedure that was quite similar to the procedure of the Brooks and Bouton (1994) study in which they demonstrated that the cue was a retrieval cue that activated the memory-representation of the extinction context and not a direct conditioned inhibitor nor an additional part of the context, it is still possible that these other mechanisms are at play in this study. At this point, we do not even know whether the difference between the groups is caused by an enhanced renewal effect in group AC or an impaired renewal effect in group EC. The results of Vansteenwegen et al. (2005), however, suggest that the latter possibility is more plausible. Especially the skin conductance data support this conclusion: The contextual renewal effect observed in Vansteenwegen et al. (2005) is completely absent in group EC of the present experiment. Moreover, the size of the renewed response in group AC is not larger (actually somewhat smaller!) than in the renewal group of Vansteenwegen et al. (2005).[1] These observations suggest that the difference between groups AC and EC is mainly due to the cue having a thwarting effect on renewal in group EC. However, additional research is certainly needed to decide on these matters.

Apart from the theoretical interest, the interaction between cues and contexts is also relevant within a clinical framework. Extinction of conditioned fear is viewed as analogous to the clinical treatment of anxiety disorders, which often involves repeated confrontations with the fear-eliciting object. Contextual renewal is taken as one of the mechanisms that can underly cases of relapse after successful treatment (Bouton, 2002). Although it may ask some creativity from the therapist to find appropriate retrieval cues (mental cues as used in the study of Mystkowski et al. (in press) are perhaps most promising), the use of retrieval cues to influence the renewal effect embodies a promising tool for the prevention of relapse. The results of the present experiment help to bridge the gap between findings from animal conditioning preparations and the clinical setting.

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[1]At first sight, a comparison of the US-expectancy data at test seems to suggest the opposite. However, when we take into consideration the magnitude of the acquisition effect in the US-expectancy ratings of the two experiments, the differentiation at test in the AC-group is smaller than the acquisition differentiation, whereas the differentiation at test in the Vansteenwegen et al. (2005) study is comparable to the acquisition differentiation.

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